

8.3.4. Design and Development Controls

The organization is required to apply controls to the design and development process



Retain documented information on design and development control activities

- ✓ Defined outcomes including specifications, design intent, functional and performance requirements, customer/end user expectations
 - ✓ Design review process with functional representation from the customer, engineering, production, quality, project management etc.), design review gates (e.g. preliminary design review, detail design review, critical design review), commercial/technical considerations, authorized progression to next stage;
 - ✓ Verification activities such as modelling, simulations, alternative calculations, comparison with other proven designs, experiments, tests, and specialist technical reviews;
 - ✓ Validation activities such as functional testing, performance testing, trials, prototypes, demonstrations, and simulations;
 - ✓ Management of actions arising from design reviews, verification or validation activities e.g. action registers, ownership, timescales, escalation, changes to risk profile.
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Auditing Design and Development Review

- ✓ Do reviews occur at planned stages throughout the design process?
- ✓ Are the reviews carried out in a systematic way involving representatives of the functions concerned with the stage(s) being reviewed?
- ✓ Have all original and any new inputs been considered ?
- ✓ Have revised inputs and outputs been reviewed and approved by those with the relevant responsibility and authority?
- ✓ Does the output demonstrate the suitability, adequacy and effectiveness of the designed product or service?
- ✓ Are there adequate records of reviews?

Auditing Design and Development Verification

- ✓ Required verifications are planned and that verification is performed as appropriate during the design and development process
- ✓ The completed design or development is acceptable and the results are consistent with and traceable to the initial requirements
- ✓ The completed design or development is the result of implementation of a proper sequence of events, inputs, outputs, interfaces, logic flow, allocation of timing, etc.
- ✓ The design or development provides safety, security, and compliance with other requirements and design inputs;
- ✓ Evidence is available to demonstrate that the verification results and any further actions have been recorded and confirmed when actions are completed.

Auditing Design and Development Validation

- ✓ Ensure there are records to confirm that the validations have been carried out
- ✓ The validation was carried out in accordance with the planned arrangements for validation
- ✓ The validation indicates that the resulting product or service is capable of meeting the requirements of the specification
- ✓ Ensure there are records of any actions necessary to correct non-compliance with the design and development inputs and the reasons for these deviations.